Stieltjes moment classification of Apéry polynomials

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Abstract

We prove that the Apéry polynomials $A_n(x) = \sum_{k=0}^{n} \binom{n}{k}^2 \binom{n+k}{k}^2 x^k$ form a Stieltjes moment sequence in n, for real $x \geq 0$, exactly when $x = 0$ or $x \geq 1$. This settles Sokal's conjecture for real parameters. Both directions use one representation of $A_n(x)$ as Legendre moments of an explicit angular probability law. For $0 < x < 1$, a Stieltjes representation would force part of this law to have an analytic density, and a support gap, an interior saddle or a degenerate endpoint rules that out. For $x > 1$, the law has an elliptic density whose continuation has a positive jump across the real cut. Explicit branch formulas and a uniform large-circle bound give a Cauchy representation, and Legendre functions of the second kind turn it into a positive measure on $[0, 1]$. The case $x = 1$ is Edgar's theorem. We also give exact negative Hankel-minor certificates for $0 < x \leq 2/3$.

Keywords: Apéry polynomials; Stieltjes moment problem; Legendre polynomials; elliptic integrals; Hankel matrices.

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1 Introduction

A real sequence $(a_n)_{n\geq 0}$ is a *Stieltjes moment sequence* if $a_n = \int_0^\infty t^n \, d\nu(t)$ for a positive Borel measure ν. Define

$$A_n(x) = \sum_{k=0}^{n} \binom{n}{k}^2 \binom{n+k}{k}^2 x^k. \tag{1}$$

Wang and Zhu posed moment questions for generalized Apéry polynomials, including the family (1) [1, Section 5]. Sokal's conjecture specifies the Stieltjes property for $x \geq 1$ and its failure for $0 < x < 1$ [2, slide 29]. Edgar proved the case $x = 1$ [3]. We prove the full classification for real parameters.

Theorem 1. *For real $x \geq 0$, $(A_n(x))_{n\geq 0}$ is a Stieltjes moment sequence if and only if $x = 0$ or $x \geq 1$.*

The stronger conjecture that $(A_n(1+x))_{n\geq 0}$ has coefficientwise totally positive Hankel matrices remains open.

1.1 Proof overview

Both parameter ranges are treated through an identity $A_n(x) = \int P_n(y)\, d\mu_r(y)$, where $r = \sqrt{x}$ and μ_r is an explicit compactly supported probability law. Its part on $[1, \infty)$ already gives positive power moments by Laplace's integral for P_n. The question is what happens to the part on $[-1, 1)$.

For $0 < r < 1$, the Legendre Poisson kernel shows that a Stieltjes representation would force this restricted law to have an analytic density. Its geometry contradicts that requirement: there is a gap in the support for $r < 1/2$, a saddle singularity for $r > 1/2$, and a quartic endpoint at $r = 1/2$. For $r > 1$, we instead express the density by a complete elliptic integral. We continue it to the slit plane, compute its positive boundary jump, and recover it by Cauchy inversion. The classical positive Q_n transform then converts the remaining Legendre moments into power moments on $[0, 1]$.

1.2 Related work

Common showed that Legendre series with Stieltjes coefficients continue to a cut plane, and that positive mixtures of the functions Q_n give Stieltjes moment sequences [4, CERN preprint, Theorems 2–3 and 5]; he notes that Yndurain found the second result independently. We include the finite-measure and endpoint details needed in our setting. The new ingredients are the geometric obstruction for $0 < x < 1$ and the positive elliptic boundary calculation for $x > 1$.

At $x = 1$, our angular representation is the formula attributed to Lupaş in 2005 and proved by Ivanisvili in 2014 [5]. The underlying product and generating identities are classical. Matala-Aho derived a recurrence for the same polynomial family [6, Theorem 2], and Neuschel studied its asymptotics and zero distributions [7]; neither is needed here. Zhu and Sun studied arithmetic properties of scalar Apéry Hankel determinants [8]. Section 6 adds finite sign certificates for a range of parameters.

2 A regularity obstruction

Let P_n denote the Legendre polynomial normalized by $P_n(1) = 1$. We first isolate the needed consequence of the classical Poisson kernel.

Lemma 2. *Let μ be a finite positive measure supported on $[-1, H]$, where $H \geq 1$. If $a_n = \int P_n(y)\, d\mu(y)$ is a Stieltjes moment sequence, then $\mu|_{[-1,1)}$ has a density real analytic on $(-1, 1)$ and bounded in a right neighborhood of -1.*

Proof. For $y \geq 1$, Laplace's integral representation gives

$$P_n(y) = \frac{1}{\pi} \int_0^\pi \left(y + \sqrt{y^2 - 1}\cos\phi\right)^n d\phi. \tag{2}$$

Hence the contribution from $[1, H]$ is the moment sequence of a finite positive measure ν_+ on $[0, R]$, where we may take $R = \max\{2, H + \sqrt{H^2 - 1}\}$. Since $|P_n(y)| \leq 1$ on $[-1, 1]$ and $|P_n(y)| \leq R^n$ on $[1, H]$, we have $|a_n| \leq \mu([-1, H])R^n$. Any positive Stieltjes measure ν representing (a_n) must therefore be supported on $[0, R]$: positive mass beyond $R + \delta$, for any $\delta > 0$, would contradict this exponential bound.

The finite signed measure $\sigma = \nu - \nu_+$ has moments

$$b_n = \int_{[-1,1)} P_n(y)\, d\mu(y), \qquad |b_n| \leq \mu([-1, 1)). \tag{3}$$

These bounds force σ to be supported on $[0,1]$, even though σ need not be positive. Indeed, its Cauchy transform agrees, for $|z| > R$, with $\sum_{n\geq 0} b_n z^{-n-1}$. The latter series is holomorphic on $|z| > 1$. The identity theorem identifies it with the actual Cauchy transform in the upper half-plane where $|z| > 1$. This continuation is real on $(1,\infty)$, so the signed Stieltjes inversion formula makes σ vanish there. The inversion follows by integrating the Poisson kernel and applying dominated convergence against the total-variation measure; intervals with non-atomic endpoints suffice to exclude all mass in $(1,\infty)$.

For $0 \leq s < 1$ put

$$K(s,y) = \frac{1-s^2}{2(1-2sy+s^2)^{3/2}}, \qquad -1 \leq y \leq 1. \tag{4}$$

The generating function for P_n, followed by differentiation and orthogonality, gives

$$K(s,y) = \frac{1}{2}\sum_{n\geq 0}(2n+1)s^n P_n(y), \qquad \int_{-1}^{1} P_n(y)K(s,y)\,dy = s^n.$$

In particular, $K(s,y)\,dy$ is a probability measure. For $s = 1$ use δ_1 instead. Integrating these measures against σ produces a signed measure η of total variation at most $|\sigma|([0,1])$. Fubini's theorem applies because $|P_n| \leq 1$ on $[-1,1]$, and gives its Legendre moments as b_n. Since the Legendre polynomials span all polynomials, uniqueness of finite signed measures on this compact interval implies $\eta = \mu|_{[-1,1)}$.

Away from $y = 1$, its density is $\int_{[0,1)} K(s,y)\,d\sigma(s)$; the possible atom $\sigma(\{1\})\delta_1$ is separate. For each real $y_0 < 1$, $1 - 2sy_0 + s^2$ has a positive minimum over $0 \leq s \leq 1$. Thus the kernels are holomorphic and uniformly bounded in a small complex neighborhood of y_0. Integration against the finite signed measure preserves holomorphy. This proves analyticity on $(-1,1)$ and, by taking $y_0 = -1$, boundedness near the left endpoint. □

3 The angular representation

The classical identities used in this section follow from the Legendre addition theorem and integral representations; see [9]. The finite expansion

$$P_n(1+2z) = \sum_{k=0}^{n} \binom{n}{k}\binom{n+k}{k} z^k$$

and Parseval's identity imply, with $r = \sqrt{x}$,

$$A_n(x) = \mathbb{E}_t \left|P_n(1+2re^{it})\right|^2, \tag{5}$$

where t is uniform on $[0, 2\pi]$. The Legendre product formula is

$$P_n(a)P_n(b) = \frac{1}{\pi}\int_0^{\pi} P_n\left(ab + \sqrt{a^2-1}\sqrt{b^2-1}\cos\theta\right)d\theta. \tag{6}$$

Although initially stated for real arguments, this is a polynomial identity in a, b: after integration, only even powers of the product of square roots remain. It therefore extends to complex arguments and is independent of the choices of square roots. For $b = \overline{a}$, choose conjugate square roots. Equations (5)–(6) yield

$$A_n(x) = \mathbb{E}P_n(Y_r), \tag{7}$$

where t, θ are independent and uniform on $[0, 2\pi]$, and

$$Y_r = 1 + 4r^2 + 4r\cos t + 4r\sqrt{1 + 2r\cos t + r^2}\cos\theta. \tag{8}$$

Using the full circle for θ does not change its cosine distribution. Writing $a = 1 + 2re^{it}$, we have $Y_r = |a|^2 + |a^2 - 1|\cos\theta$. Hence the triangle inequality gives

$$-1 \le Y_r \le 1 + 8r^2 + 8r. \tag{9}$$

For $0 < r < 1$, the square root in (8) is strictly positive and smooth.

For completeness, when $r = 1$, substituting $t = 2\alpha$ in (8) gives the same distribution as $1 + 8\cos^2\alpha + 8\cos\alpha\sin\beta$ with independent uniform angles: the absolute value of $\cos\alpha$ is absorbed by symmetry in the second angle. This is precisely the earlier Lupaş scalar formula [5]. We do not use the $r = 1$ case in the proof.

4 Failure on $0 < x < 1$

Theorem 3. *For every fixed real x with $0 < x < 1$, the sequence $(A_n(x))_{n\ge 0}$ is not a Stieltjes moment sequence.*

Proof of Theorem 3. Fix $0 < r < 1$, and suppose (7) is a Stieltjes moment sequence. Lemma 2 applies to the distribution μ_r of Y_r. Set

$$h = 1 + 8r^2 - 8r.$$

At $(t, \theta) = (\pi, \pi)$, $Y_r = h < 1$, so by continuity the law has positive mass below 1. We distinguish three cases.

Case 1: $0 < r < 1/2$**.** For $c = \cos t$, the lower endpoint as θ varies is

$$f(c) = 1 + 4r^2 + 4rc - 4r\sqrt{1 + 2rc + r^2}.$$

Since

$$f'(c) = 4r\left(1 - \frac{r}{\sqrt{1 + 2rc + r^2}}\right) > 0, \qquad -1 \le c \le 1,$$

where $\sqrt{1 + 2rc + r^2} \ge 1 - r > r$, the global lower endpoint is $f(-1) = h \in (-1, 1)$. Thus μ_r vanishes on $(-1, h)$ but has positive mass in $(h, 1)$. The analytic density furnished by Lemma 2 would vanish identically on the connected interval $(-1, 1)$, a contradiction.

Case 2: $1/2 < r < 1$**.** Let $u = t - \pi$ and $v = \theta - \pi$. Expansion at the interior point of the angular torus gives

$$Y_r = h + \frac{2r(1 - 2r)}{1 - r}u^2 + 2r(1 - r)v^2 + O\big((|u| + |v|)^4\big). \tag{10}$$

The two quadratic coefficients are nonzero and have opposite signs. The Morse lemma supplies local smooth coordinates, with a bounded positive Jacobian, in which $Y_r - h = U^2 - V^2$. After a further invertible linear change, this is the product of the coordinates. In a fixed small square the region where the absolute value of that product is at most ϵ has area at least $c\epsilon\log(1/\epsilon)$ for all sufficiently small $\epsilon > 0$. The original angular probability density is a positive constant. Consequently

$$\mu_r([h - \epsilon, h + \epsilon]) \ge c'\epsilon\log(1/\epsilon).$$

Since $h \in (-1,1)$, this contradicts the $O(\epsilon)$ bound required by a density bounded near h.

Case 3: $r = 1/2$**.** Now $h = -1$ and direct expansion gives

$$Y_{1/2} = -1 + \tfrac{1}{2}(u^4 + v^2) + O(u^6 + u^2v^2 + v^4). \tag{11}$$

Together with (9), this implies $0 \le Y_{1/2} + 1 \le C(u^4 + v^2)$ locally. For a sufficiently small fixed $c > 0$, the rectangle

$$|u| \le c\epsilon^{1/4}, \qquad |v| \le c\epsilon^{1/2}$$

is mapped into $[-1, -1+\epsilon]$. It has probability at least $c'\epsilon^{3/4}$. This contradicts the $O(\epsilon)$ mass bound from the density bounded near -1 in Lemma 2. The three cases cover $0 < x = r^2 < 1$. □

5 A positive elliptic kernel for $x > 1$

Theorem 4. *For every real $x > 1$, the sequence $(A_n(x))_{n\ge 0}$ has a positive compactly supported Stieltjes representing measure.*

Fix $r = \sqrt{x} > 1$ and set

$$h = 1 + 8r^2 - 8r, \qquad H = 1 + 8r^2 + 8r, \qquad c_r = \frac{1}{2\pi^2 r}.$$

Write f_r for the density of the Legendre averaging law on $(-1,1)$. We will continue f_r to $\Omega = \mathbb{C} \setminus [1,\infty)$, express it as the Cauchy transform of a positive density, and convert its Legendre moments into power moments.

5.1 The density and its continuation

Throughout this section $\mathcal{K}$ uses a *parameter*, not a modulus:

$$\mathcal{K}(w) = \int_0^{\pi/2} (1 - w\sin^2\theta)^{-1/2}\, d\theta = R_F(0, 1-w, 1).$$

Here R_F is Carlson's symmetric integral, continued from

$$R_F(a,b,c) = \frac{1}{2}\int_0^\infty \frac{dt}{\sqrt{(t+a)(t+b)(t+c)}}.$$

Each factor uses its principal square root on the initial domain. These conventions agree with [9, Sections 19.16 and 19.25] after replacing its squared modulus by our parameter.

For $-1 < y < 1$, put

$$d = \sqrt{(y+1)/2}, \quad m = \frac{y-1}{4r}, \quad A = (1-d)^2 - m^2, \quad B = (1+d)^2 - m^2. \tag{12}$$

Conditional on $c = \cos t$, the second angular variable has an arcsine law. Its squared radicand factors as

$$16r^2(1 + 2rc + r^2) - (y - 1 - 4r^2 - 4rc)^2 = 16r^2\big[d^2 - (c-m)^2\big].$$

The conditional law has no atoms, since its amplitude is at least $4r(r-1) > 0$. Integrating against the arcsine density of c gives

$$f_r(y) = \frac{1}{4\pi^2 r}\int_{m-d}^{m+d} \frac{dc}{\sqrt{(1-c^2)(c-m+d)(m+d-c)}}.$$

Indeed, the roots of the conditional quadratic radicand are $m-d, m+d$, both strictly inside $(-1,1)$ when $r>1$ and $-1<y<1$. For example, $1+m-d=(1-d)(1-(d+1)/(2r))>0$ and $1-m-d=(1-d)(1+(d+1)/(2r))>0$. The four-root substitution

$$t^2=\frac{(e_3-e_1)(c-e_2)}{(e_3-e_2)(c-e_1)},\qquad (e_1,e_2,e_3,e_4)=(-1,m-d,m+d,1),$$

reduces this integral to the standard complete elliptic integral. Since $(e_3-e_1)(e_4-e_2)=B$ and $(e_3-e_2)(e_4-e_1)=4d$, it yields

$$f_r(y)=c_rR_F(0,A,B)=c_rB^{-1/2}\mathcal{K}(4d/B). \tag{13}$$

Lemma 5. *The germ in (13) extends to a single-valued holomorphic function on Ω.*

Proof. On the two-sheeted surface $d^2=(z+1)/2$, the finite singularities of the composition in (13) can occur only above $-1,1,h,H$. Indeed, $\mathcal{K}(p)=\pi\,{}_2F_1(1/2,1/2;1;p)/2$ continues along every path avoiding $p=0,1,\infty$, by the classical hypergeometric equation [9, 15.10.1]. Here $p=4d/B$, and the preimages are $d=0$, $A=0$, or $B=0$. The factorizations

$$A=(d-1)^2\left(1-\frac{(d+1)^2}{4r^2}\right),\qquad B=(d+1)^2\left(1-\frac{(d-1)^2}{4r^2}\right) \tag{14}$$

give precisely those projected points. Critical points of the parameter map create no singularity in a composition, and the prefactor square root continues along paths avoiding its zeros. Thus the selected germ continues along every path in $\Omega\setminus\{-1\}$.

The apparent interior puncture is removable for this germ. The change $c=m+d\cos\phi$ in the density integral gives, near $d=0$,

$$f_r=\frac{1}{4\pi^2 r}\int_0^\pi\left[1-\left(\frac{d^2-1}{2r}+d\cos\phi\right)^2\right]^{-1/2}d\phi.$$

At $d=0$ the expression inside the inner square has modulus $1/(2r)<1$. For sufficiently small complex d the integrand has a single holomorphic square-root branch uniformly in ϕ. Its integral is holomorphic in d, and $\phi\mapsto\pi-\phi$ shows it is even. It is therefore holomorphic in $z=2d^2-1$ and agrees with the initial germ for real $d>0$. The generator loop about -1 has trivial monodromy. Since Ω is simply connected and the fundamental group of $\Omega\setminus\{-1\}$ is generated by that loop, the monodromy theorem gives single-valuedness. The local integral then fills the puncture. □

5.2 The boundary values

The parameter versions of the classical connection identities are

$$\mathcal{K}(z)=(1-z)^{-1/2}\mathcal{K}\big(z/(z-1)\big),\qquad z<0, \tag{15}$$

$$\mathcal{K}(z)=z^{-1/2}\big[\mathcal{K}(1/z)+i\mathcal{K}(1-1/z)\big],\qquad z \text{ above } (1,\infty). \tag{16}$$

They hold as identities of the adjacent holomorphic germs; the lower-bank version of (16) is its conjugate. See [9, 19.7.2–19.7.3], including its explicit upper/lower sign rule. The convergent logarithmic expansion [9, 19.12.1] gives

$$\mathcal{K}(1-\epsilon)=-\frac{\mathcal{K}(\epsilon)}{\pi}\log\epsilon+G(\epsilon), \tag{17}$$

where G is holomorphic near zero.

For real $y>1$, retain the positive square root in d. The factorizations (14) show that A has a double zero at 1 and a simple zero at h, while B has a simple zero at H.

Lemma 6. *The upper boundary value of f_r is given by the following holomorphic germs adjacent to the indicated real intervals:*

$$f_r(z) = \frac{c_r}{\sqrt{B}}\big[\mathcal{K}(p) + 2i\mathcal{K}(1-p)\big], \qquad p = \frac{4d}{B}, \quad 1 < y < h, \tag{18}$$

$$f_r(z) = \frac{c_r}{\sqrt{4d}}\big[\mathcal{K}(q) + 3i\mathcal{K}(1-q)\big], \qquad q = \frac{B}{4d}, \quad h < y < H, \tag{19}$$

$$f_r(z) = \frac{c_r}{\sqrt{-A}}\big[-2\mathcal{K}(t) + 3i\mathcal{K}(1-t)\big], \qquad t = \frac{B}{A}, \quad H < y < \infty. \tag{20}$$

The parameters p, q, t respectively lie in $(0,1)$ on their displayed real intervals, and all displayed real prefactors use positive roots.

Proof. Near 1,

$$A = \frac{1-r^{-2}}{16}(z-1)^2 + O((z-1)^3), \qquad B = 4 + O(z-1).$$

Going around 1 through the upper half-plane makes A/B wind clockwise once about zero. Equation (17) adds exactly $2i\mathcal{K}(A/B)$, proving (18) near 1. On compact subintervals of $(1,h)$, both p and $1-p$ stay in $(0,1)$. Overlapping neighborhoods therefore propagate the same germ without an additional period.

At h,

$$A'(h) = -\frac{(r-1)^2}{r(2r-1)} < 0, \qquad p'(h) = -A'(h)/B(h) > 0.$$

An upper semicircle about h thus sends p through the upper side of 1. Apply (16) to $\mathcal{K}(p)$ and (15) to $\mathcal{K}(1-p)$:

$$\mathcal{K}(1-p) = p^{-1/2}\mathcal{K}(1-1/p).$$

Substitution in (18) gives (19), including the coefficient 3. The square-root factors agree on these germs because $Bp = 4d$ and they agree before crossing. Again, compact subintervals of (h,H) keep $q, 1-q$ strictly between 0 and 1.

At H,

$$B'(H) = -\frac{(r+1)^2}{r(2r+1)} < 0, \qquad q'(H) = B'(H)/(4d(H)) < 0.$$

Hence q crosses zero through the lower half-plane and $1-q$ crosses 1 through the upper half-plane. Apply (15) to $\mathcal{K}(q)$ and (16) to $\mathcal{K}(1-q)$. With

$$\frac{q}{q-1} = t, \qquad \frac{1}{1-q} = 1-t, \qquad (4d)(1-q) = -A,$$

the real coefficient becomes $1-3 = -2$, giving (20). The parameters $t, 1-t$ remain in $(0,1)$ to the right of H. Overlapping neighborhoods propagate this germ to every finite point of that bank. These are valid continuations of the selected germ in Ω, where Lemma 5 supplies single-valuedness. □

Define $\rho_r(y) = \operatorname{Im} f_r(y+i0)/\pi$. Lemma 6 gives the explicitly positive density

$$\rho_r(y) = \frac{1}{2\pi^3 r}\begin{cases} 2R_F(0,4d,B), & 1 < y < h, \\ 3R_F(0,4d,B), & h < y < H, \\ 3R_F(0,-A,-B), & H < y < \infty. \end{cases} \tag{21}$$

Thus ρ_r has a finite right limit at 1, a finite jump at h, and at most a logarithmic singularity at H. At infinity, $-A, -B \sim y^2/(16r^2)$, so homogeneity gives $\rho_r(y) = O(1/y)$.

5.3 The Cauchy representation

Lemma 7. *For every $z \in \Omega$,*

$$f_r(z) = \int_1^\infty \frac{\rho_r(a)}{a - z}\, da. \tag{22}$$

Proof. The integral converges by the local logarithmic bounds and $\rho_r(a) = O(1/a)$. To apply Cauchy's theorem, we need a bound that is uniform around a large circle, including angles close to the cut.

Take $R_0 = 1000r^2$ and the connected upper exterior domain $D = \{z : \operatorname{Im} z > 0,\ |z| > R_0\}$. Choose $d = \sqrt{(z+1)/2}$ by continuation from the positive bank. The expansion

$$A = -\frac{z^2}{16r^2} + \left(\frac{1}{2} + \frac{1}{8r^2}\right) z + \frac{3}{2} - \frac{1}{16r^2} - 2\sqrt{(z+1)/2}$$

gives $|A + z^2/(16r^2)| \le 5R/4$, $R = |z| > R_0$, at every angle. Thus

$$\frac{R^2}{32r^2} \le |A| \le \frac{3R^2}{32r^2}, \qquad \frac{\sqrt{R}}{2} \le |d| \le \sqrt{R}.$$

Put $w = 1 - t = -4d/A$ and $A = -z^2(1+e)/(16r^2)$. Then

$$\frac{64r^2}{3R^{3/2}} \le |w| \le \frac{128r^2}{R^{3/2}} < \frac{1}{2}, \qquad |e| \le \frac{1}{50}.$$

Use $0 < \arg z < \pi$ and the small analytic logarithms of $1 + 1/z$ and $1 + e$. The logarithm of w continued from the positive bank is

$$\operatorname{Log} w = \log \frac{64r^2}{\sqrt{2}} - \frac{3}{2} \operatorname{Log} z + \frac{1}{2} \log(1 + 1/z) - \log(1 + e). \tag{23}$$

In particular,

$$|\operatorname{Im} \operatorname{Log} w| \le \frac{3\pi}{2} + \tfrac{1}{2} \arcsin(1/1000) + \arcsin(1/50) < 5, \qquad |\operatorname{Log} w| \le \tfrac{3}{2} \log R + 5.$$

Here $R^{-3/2} < |w| < 1$ controls the real part. This logarithm is *continued*, not replaced by the principal logarithm: its imaginary part can approach $-3\pi/2$ near the negative real axis. The square root of $-A$ on D is chosen positive at the positive bank; its modulus is $|A|^{1/2}$.

In (17), the convergent series [9, 19.12.1–19.12.3] is

$$G(w) = \sum_{j \ge 0} \left(\frac{(1/2)_j}{j!}\right)^2 [\psi(j+1) - \psi(j+1/2)] w^j.$$

Its coefficients satisfy $((1/2)_j/j!)^2 \le 1$ and $0 < \psi(j+1) - \psi(j+1/2) \le 2\log 2$. Hence for $|w| \le 1/2$,

$$|\mathcal{K}(w)| \le \pi/\sqrt{2}, \qquad |G(w)| \le 4 \log 2.$$

Equation (20) becomes

$$f_r(z) = \frac{c_r}{\sqrt{-A}} \left[\frac{2}{\pi} \mathcal{K}(w) \operatorname{Log} w - 2G(w) + 3i\mathcal{K}(w)\right]. \tag{24}$$

All terms on the right are holomorphic on D with the specified branches. The identity theorem extends this identity from its positive-bank neighborhood to all D. It gives the explicit full-semicircle bound

$$|f_r(z)| \le \frac{\sqrt{32}}{2\pi^2 R}\left[\sqrt{2}\left(\tfrac{3}{2}\log R + 5\right) + 8\log 2 + \frac{3\pi}{\sqrt{2}}\right].$$

Schwarz reflection gives the lower semicircle, and holomorphy supplies the negative-axis limit. Thus the estimate is uniform on the entire slit circle, and the large-circle Cauchy integral vanishes.

At 1, A/B has a double zero and (17) gives $f_r(z) = O(1 + |\log|z-1||)$. At h, $1-p$ has a simple zero in (18); at H, q has a simple zero in (19). The same logarithmic expansion therefore gives $f_r(z) = O(1 + |\log|z-a||)$ near each $a \in \{1, h, H\}$ on either side of the cut. An arc of radius ϵ contributes $O(\epsilon(1 + |\log\epsilon|))$, which tends to zero. Cauchy's theorem on the slit plane now gives (22), with $(f_+ - f_-)/(2\pi i) = \rho_r$ and denominator $a - z$. □

5.4 The positive remainder measure

Common's positive Froissart–Gribov theorem [4, CERN preprint, Section 4, Theorem 5] converts positive Q_n mixtures into Stieltjes moments. We give the integral calculation for the present density, whose spectral support reaches 1 and whose total spectral mass need not be finite.

Use the standard normalization

$$Q_n(a) = \frac{1}{2}\int_{-1}^{1} \frac{P_n(y)}{a-y}\,dy, \qquad a > 1.$$

Heine's integral is

$$Q_n(a) = \int_0^\infty \left(a + \sqrt{a^2-1}\cosh u\right)^{-n-1} du.$$

Set $t = (a + \sqrt{a^2-1}\cosh u)^{-1}$. Since $du = -dt/(t\sqrt{1-2at+t^2})$, this gives

$$Q_n(a) = \int_0^{a-\sqrt{a^2-1}} \frac{t^n}{\sqrt{1-2at+t^2}}\,dt. \tag{25}$$

In [9, 14.12.13–14.12.14], Olver's normalized function is $Q_n/n!$; multiplying both formulas by $n!$ yields precisely this normalization.

Proof of Theorem 4. First take $n = 0$. All kernels in (22) and (25) are nonnegative on their real integration domains, so Tonelli's theorem gives

$$\int_{-1}^{1} f_r(y)\,dy = 2\int_1^\infty \rho_r(a)Q_0(a)\,da = \int_0^1 w_r(t)\,dt < \infty,$$

where w_r is defined below. The first integral is the probability $\mu_r([-1,1))$; the endpoints have zero mass by the conditional arcsine representation. For general n, put $b_n = \int_{-1}^{1} P_n(y)f_r(y)\,dy$. The bounds $|P_n(y)| \le 1$ and $0 \le t^n \le 1$ reduce absolute integrability to the case $n = 0$. Fubini's theorem now gives

$$b_n = 2\int_1^\infty \rho_r(a)Q_n(a)\,da = \int_0^1 t^n w_r(t)\,dt, \tag{26}$$

$$w_r(t) = 2\int_1^{(t+t^{-1})/2} \frac{\rho_r(a)}{\sqrt{1-2at+t^2}}\,da \ge 0, \qquad 0 < t < 1. \tag{27}$$

The domain in (27) follows from $t \le a - \sqrt{a^2-1}$ if and only if $1 \le a \le (t+t^{-1})/2$. Thus $w_r(t)\,dt$ is a finite positive measure on $[0,1]$ with moments b_n.

The contribution from $Y_r \ge 1$ already has the positive Laplace-mixture measure ν_+, supported on $[H-\sqrt{H^2-1}, H+\sqrt{H^2-1}]$. Adding the positive measure in (27) proves that $\nu_+ + w_r(t)\,dt$ is a positive compact Stieltjes measure with moments $A_n(x)$. □

Proof of Theorem 1. Theorem 3 excludes $0 < x < 1$ and Theorem 4 covers $x > 1$. Edgar's prior theorem [3] covers $x = 1$. At $x = 0$, $A_n(0) = 1$ is represented by the point mass at 1. □

6 Hankel certificates and fixed-size tests

Zhu and Sun studied integrality and divisibility of scalar Apéry Hankel determinants [8]. Here we give finite sign certificates for an interval of parameters. Define

$$H_{d,s}(x) = \det\big(A_{i+j+s}(x)\big)_{0\le i,j<d}, \qquad s \in \{0,1\}.$$

Proposition 8. *For every $0 < x \le 2/3$, at least one of $H_{4,0}(x), H_{8,0}(x), H_{9,1}(x), H_{10,0}(x)$ is strictly negative.*

Proof. Table 1 gives exact sign certificates. For each row, $Q(x) = H_{d,s}(x)/x^v$ is a polynomial of degree N.

Table 1: Intervals and polynomial degrees for the negative-minor certificates.

(d,s)	a	b	v	N
$(4,0)$	0	$9/100$	3	9
$(8,0)$	$1/20$	$101/500$	14	42
$(9,1)$	$201/1000$	$47/100$	18	63
$(10,0)$	$41/100$	$2/3$	23	67

Writing $Q(a+(b-a)t) = \sum_{j=0}^{N} c_j t^j$, its Bernstein coefficients are

$$\beta_k = \sum_{j=0}^{k} c_j \frac{\binom{k}{j}}{\binom{N}{j}}, \qquad 0 \le k \le N.$$

In each of the four rows, every β_k is a strictly negative rational number. The supplemental material records all coefficients; its verifier recomputes the determinants over $\mathbb{Q}[x]$, verifies their valuations and degrees, checks each rational sign, and reconstructs

$$Q(a+(b-a)t) = \sum_{k=0}^{N} \beta_k \binom{N}{k} t^k (1-t)^{N-k}.$$

The Bernstein basis is nonnegative and sums to one on $[0,1]$, proving $Q < 0$ throughout $[a,b]$. Hence $H_{d,s} < 0$ on that interval wherever $x > 0$. The four intervals overlap and cover $(0, 2/3]$. □

6.1 Why a fixed finite test fails near 1

No fixed finite set of Hankel minors detects every parameter in $(0,1)$. Indeed, Edgar's theorem [3, Theorem 1] provides a nonzero positive integrable moment density at $x=1$. For any fixed d and $s\in\{0,1\}$, and any nonzero real vector v,

$$\sum_{i,j=0}^{d-1} v_i v_j A_{i+j+s}(1) = \int_0^\infty t^s \left(\sum_{i=0}^{d-1} v_i t^i\right)^2 d\nu(t) > 0.$$

A nonzero polynomial has finitely many zeros, whereas the absolutely continuous measure cannot be supported there. Thus each fixed Hankel matrix is positive definite at $x=1$, and by continuity any finite collection of them, or of their principal submatrices, stays positive definite in a neighborhood of 1. At each $x\in(0,1)$, Theorem 3 gives a negative principal minor of the ordinary or shifted Hankel matrix, but it does not say which minor or bound its size.

6.2 Why a large shift does not detect the threshold

Proposition 9. *For every fixed $x>0$ and positive integer d, the matrix $(A_{N+i+j}(x))_{0\le i,j<d}$ is positive definite for all sufficiently large integers N.*

Proof. Use (7) and the Laplace representation (2) to write $A_n(x)=a_n+b_n$, where $a_n=\int t^n\,d\nu_+(t)$ is the positive moment contribution from $Y\ge 1$, and $|b_n|\le 1$ is the contribution from $-1\le Y<1$. Put $r=\sqrt{x}$, $H=1+8r^2+8r>1$ and $R=H+\sqrt{H^2-1}>1$. The law of Y has positive mass arbitrarily close to its maximum H, attained at $t=\theta=0$ in (8). In the Laplace representation, varying its independent cosine angle gives ν_+ infinite support arbitrarily close to R. Choose $1<c<R$ so that $\nu_+|_{[c,R]}$ has infinite support. Its degree-$(d-1)$ Gram matrix

$$G=\left(\int_{[c,R]} t^{i+j}\,d\nu_+(t)\right)_{0\le i,j<d}$$

is strictly positive definite, since a nonzero polynomial cannot vanish on that support. Write $\lambda=\lambda_{\min}(G)>0$. The matrix (a_{N+i+j}) dominates $c^N G$ in positive-semidefinite order, whereas (b_{N+i+j}) has operator norm at most d. Hence

$$\lambda_{\min}\big((A_{N+i+j}(x))_{0\le i,j<d}\big)\ge c^N\lambda - d > 0$$

for all sufficiently large N. □

Thus fixed-size tests at increasingly large shifts become positive on both sides of the parameter threshold. The cutoff may depend on x and d.

Supplemental material

Python scripts and exact rational certificates for Proposition 8, algebraic identities used in the proof, and high-precision numerical checks of the elliptic continuation and the dispersion formula. The sign certificates use exact rational arithmetic.

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Data availability statement

The verification scripts and certificate data are provided as supplemental material. No other data were used.

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